Ritesh Chowdhry

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EDUCATION

Ph.D. in Electrical and Computer Engineering **Aug. 2021 - Present**
University of Florida
Thesis topic: Beyond Band Masking: Decomposition-Guided Self-Supervised Learning for Hyperspectral Images

M.S. in Electrical and Computer Engineering **Aug. 2019 - Aug. 2021**
University of Florida

B.S. in Electronics and Communication Engineering **Aug. 2014 - Aug. 2018**
Kurukshetra University

SKILLS

- **Languages:** Python, C++, SQL
- **Tools:** PyTorch, TensorFlow, Slurm, Linux, ROS, Git, Comet, Wandb, Jupyter Notebooks
- **Hardware:** HPC, LiDAR, Raspberry Pi, Jetson Nano, Bluetooth/Wi-Fi/GSM modules, ultrasonic, infrared and moisture sensors.

WORK EXPERIENCE

Amazon **June 2026 - Aug. 2026**
Applied Scientist Intern *Seattle, WA*
Cost- and Latency-Aware Benchmarking and Routing for Large Language Models

- Built the first public answer-key benchmark to carry all three routing quantities in the same cell, correctness, dollar cost, and served-model latency, over a pool including closed commercial flagship models: 402,510 labeled cells spanning 15 models from 11 vendors and 18 public datasets (26,834 queries).
- Established that cost and latency are separate routing objectives, showing the cheapest-correct and fastest-correct model disagree on 34.1% of solvable queries, and that a query-blind policy lands within 4.9% of optimal cost yet overspends latency by 26.6%.
- Developed a query-level router combining a nearest-neighbor capability estimate with a learned MLP head under a joint cost-latency penalty, matching the strongest single model's accuracy while cutting 20.5% of cost and 7.9% of latency on all 20 seeds.
- Re-implemented six published routing methods behind a single scoring contract for like-for-like comparison, with paired-seed bootstrap and permutation testing to separate real wins from noise.
- Added a calibrated confidence and abstention layer (calibration error 0.017) returning a per-query recommendation card with the chosen model, its odds of being correct, and an escalation flag, raising accuracy from 77.9% to 89.1% at 80% coverage.

Amazon **June 2025 - Aug. 2025**
Applied Scientist Intern *Seattle, WA*
Agentic Plot Analysis System for Financial Reports

- Developed an agentic system leveraging vision-language models (VLMs) to automate financial report analysis for product teams, reducing manual analysis time by 20-30 minutes per report across hundreds of daily/bi-daily generated scenarios.
- Engineered multimodal data extraction pipeline integrating VLMs via AWS Bedrock to process line plots, tables, and markdown reports, enabling simultaneous analysis of 5-6 interconnected business metrics.
- Designed prompt engineering and chain-of-thought reasoning workflows to generate comprehensive insights and recommendations, identifying cross-metric correlations and actionable patterns for decision-making.
- Built robust context engineering and preprocessing pipelines to standardize diverse data formats from multiple sources, ensuring consistent and reliable analytical outputs at scale.

NumFOCUS Organization **June 2025 - Aug. 2025**

Google Summer of Code Contributor *Gainesville, FL*

NEON Multi-Modal Tree Species Dataset and Pipeline Development

- Selected among 8% of applicants for Google Summer of Code program to develop open-source package NeonCrops for ecological research.
- Created comprehensive multimodal dataset of 234 tree species with 41,738 individual tree crowns from 38 NEON sites across North America, including RGB imagery, 369-band hyperspectral data, and LiDAR canopy height models.
- Developed end-to-end NEON data pipeline for all three modalities: curating shapefiles, automated downloading, cropping, and data curation to provide research-ready datasets, released on Hugging Face under an MIT license.
- Engineered preprocessing workflows and classification pipelines, enabling researchers to access multimodal tree crown data and models without extensive data processing expertise.

USDA Appalachian Fruit Research **June 2023 - Aug. 2023**

Machine Learning Intern *Kearneysville, WV (Remote)*

Segmentation Of Leafless Trees From Non-Tree Regions Using Deep Learning

- Collaborated on developing a deep learning-based semantic segmentation model (UNET) for apple orchard automation.
- Annotated a substantial image dataset for training and validation using various image annotation tools, both algorithm-assisted and manual.
- Explored post-processing techniques such as connected component analysis and morphological operations to mitigate noise from visually similar background trees.

RESEARCH EXPERIENCE

University of Florida, Machine Learning and Sensing Lab **Aug. 2021 - Present**

Doctoral Student Researcher *Gainesville, FL*

Self-Supervised Representation Learning for Hyperspectral Imagery (HSI)

- Developing novel **Masked Autoencoder (MAE)** framework using **Vision Transformers (ViT)** for HSI with component-level masking in decomposed representation spaces, addressing fundamental limitations of band-level masking in **self-supervised** learning.
- Improved pretraining efficiency by masking complete spectral components instead of random bands, creating uniform reconstruction tasks that exploit structured spectral redundancy and avoid uneven predictability issues inherent in traditional band-level masking approaches.
- Systematically evaluated 15 decomposition methods (PCA, NMF, ICA, spectral unmixing) to optimize self-supervised pretraining across multiple hyperspectral datasets.
- Achieved 15% F1 score improvement over supervised baselines on imbalanced tree species classification using self-supervised and semi-supervised learning with 2M unlabeled samples from NEON sites.

Pixel-Level Soil Water Content Prediction Using Deep Learning

- Developed first approach for pixel-level soil moisture mapping in rhizosphere systems using DenseNet-inspired CNN and hyperspectral imaging, achieving ∼0.5mm spatial resolution vs. traditional point sensors (R^2=0.89, MAE=1.58%).
- Implemented **UNet**-based semantic segmentation for root-soil separation in hyperspectral rhizobox imagery. Led creation of **HyperPRI dataset** and analyzed root spectral interference effects on soil moisture predictions.
- Designed spectral scaling correction method for domain adaptation, enabling successful transfer of CNN-based moisture prediction from controlled laboratory samples to complex plant-soil systems with pervasive root presence across 28 rhizoboxes.

Traffic Noise Analysis (UF Research AI catalyst fund) **Jan. 2021 - Aug. 2021**

Graduate Research Assistant *Gainesville, FL*

University of Florida, EAAR Lab

- Conducted comprehensive traffic noise analysis by designing and assembling two weather-resistant, real-time environmental noise recording kits, employing a Raspberry Pi, network module, and 1/2" prepolarized free-field condenser microphone.
- Collaborated with the City of Gainesville to extract traffic information from video feeds of two intersections. Leveraged this data to create a baseline **regression model** for real-time noise prediction, utilizing traffic data from the video feeds as feature vectors.

F1-tenth self driving car (based on UPenn F1tenth project) **Sept. 2019 - Dec. 2020**

Student Researcher *Gainesville, FL*
University of Florida, F1-tenth Lab

- Implemented and designed self-driving algorithms like wall following, mapping & localization, path planning using SLAM for a one-tenth model of an F1 car using a LIDAR and camera.
- Used ROS on a Linux system on an NVIDIA Jetson TX-2 (as SOC Board) for running LIDAR and stereo cameras.

PUBLICATIONS

- Graves, S. J., **Chowdhry, R.**, Zhou, M., Harmon, I., Weinstein, B., ... & Bohlman, S. A. (2025). Facilitating macrosystem biology with organismal-scale airborne remote sensing: Challenges and opportunities. Functional Ecology.
- Chang, S. J., **Chowdhry, R.**, Song, Y., ... & Zare, A. (2024). HyperPRI: A dataset of hyperspectral images for underground plant root study. Computers and Electronics in Agriculture, 225, 109307.
- Sanchit Tanwar, Ayush Tiwari, **Ritesh Chowdhry**. Indian Licence Plate Dataset in the wild (Arxiv), 2021.
  - Proposed a two-stage model that combines semantic segmentation for localization with LPRNet for character recognition, utilizing a novel annotation technique to overcome challenges posed by different plate orientations. [arXiv: 2111.06054]

KEY PROJECTS

- **Multistream Architecture for Repetetive Action Counting:** Created a multi-stream architecture, utilizing RGB (using MoviNets backbone) and pose estimation (poses extracted using YOLOv7) techniques to accurately count the number of repetitions performed during exercise routines, as well as to effectively classify the specific exercises being executed.

CONFERENCE PRESENTATIONS

- NSF Signals in the Soils Annual Workshop, 2022: Hyperspectral Signals in the Soil Poster, **R. Chowdhry**, A. Zare.

AWARDS & ACHIEVEMENTS

- Artificial Intelligence Research Catalyst Fund - UF Research (Graduate Assistantship) *Jan. 2021*
- Reached the final round of Smart India Hackathon Hardware edition *2018*

TEACHING & OUTREACH ACTIVITIES

Supervised Teaching in ECE Department at the University of Florida **Aug. 2021 - Dec. 2021**
Supervised Teacher, EEL 5840/4930 Fundamentals of Machine Learning

- Delivered a lecture on Introduction to PyTorch to 175 students
- Designed and graded course assignments and projects and held weekly office hours

Instructor in UF CPET Precollege Scholars program **July 2022**
Instructor, Survey of Machine Learning

- Designed and taught an introductory level course for high-school students.
- Created interactive notebooks to explain regression, image processing and image classification concepts.